

Multivariable Optimization

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Outline (75 minutes)

1. **Critical Points and First-Order Conditions** – Extending calculus to many variables (15 min)
2. **Second-Order Conditions and the Hessian** – Classifying critical points (15 min)
3. **Gradient Descent** – The basic optimization algorithm (10 min)
4. **Portfolio Optimization** – Mean-variance, risk-return tradeoff (20 min)
5. **Neural Network Optimization** – Loss functions and backpropagation (15 min)
6. **Practice Exercises** – Worked problems (time permitting)

1 Critical Points and First-Order Conditions

1.1 Review: One Variable

For $f(x)$, a **critical point** satisfies $f'(x) = 0$. At such a point, f may have a local max, min, or saddle (inflection).

1.2 Extension to Several Variables

Definition 1.1. A point $\mathbf{x}^* = (x_1^*, \dots, x_n^*)$ is a **critical point** of f if

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \quad \Leftrightarrow \quad \frac{\partial f}{\partial x_1} = \frac{\partial f}{\partial x_2} = \dots = \frac{\partial f}{\partial x_n} = 0$$

Why? If any partial derivative is nonzero, we could move in that direction to increase (or decrease) f . So at an extremum, all partials must vanish.

1.3 Example

For $f(x, y) = x^2 + y^2 - 2x - 4y + 5$:

$$\nabla f = (2x - 2, 2y - 4) = (0, 0) \quad \Rightarrow \quad x = 1, y = 2$$

The only critical point is $(1, 2)$. We will see it is a minimum.

2 Second-Order Conditions and the Hessian

2.1 The Hessian Matrix

Definition 2.1. The **Hessian** of $f(x, y)$ is the matrix of second partial derivatives:

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$$

For n variables, $H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

By Clairaut: $f_{xy} = f_{yx}$, so H is symmetric.

2.2 Second Derivative Test (Two Variables)

Theorem 2.2 (Second Derivative Test). Suppose (a, b) is a critical point of f and f has continuous second partials. Let

$$D = f_{xx}(a, b) \cdot f_{yy}(a, b) - (f_{xy}(a, b))^2 = \det(H).$$

- If $D > 0$ and $f_{xx}(a, b) > 0$: **local minimum**
- If $D > 0$ and $f_{xx}(a, b) < 0$: **local maximum**
- If $D < 0$: **saddle point** (neither max nor min)
- If $D = 0$: **inconclusive**

Intuition: D is the determinant of the Hessian. When $D > 0$, the Hessian is “definite” (all eigenvalues have the same sign)—either concave up (min) or concave down (max). When $D < 0$, eigenvalues have opposite signs—a saddle.

2.3 Worked Example

For $f(x, y) = x^2 + y^2 - 2x - 4y + 5$, we found the critical point $(1, 2)$.

Second partials:

$$f_{xx} = 2, \quad f_{yy} = 2, \quad f_{xy} = 0$$

So $D = 2 \cdot 2 - 0^2 = 4 > 0$ and $f_{xx} > 0$. Therefore $(1, 2)$ is a **local minimum**.

Completing the square: $f(x, y) = (x - 1)^2 + (y - 2)^2$, so the minimum is 0 at $(1, 2)$.

3 Gradient Descent

3.1 The Idea

When we cannot solve $\nabla f = \mathbf{0}$ analytically, we **iterate**: start at a point, move in the direction that decreases f (for minimization), and repeat.

Steepest descent: Move in the direction of $-\nabla f$ (opposite to the gradient, since the gradient points uphill).

3.2 The Algorithm

1. Choose initial point $\mathbf{x}^{(0)}$ and step size (learning rate) $\alpha > 0$.
2. For $k = 0, 1, 2, \dots$:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha \nabla f(\mathbf{x}^{(k)})$$

3. Stop when $|\nabla f|$ is small or after a fixed number of iterations.

Why it works: The update $\mathbf{x} - \alpha \nabla f$ moves in the direction of steepest descent. For small α , f decreases (provided we are not at a minimum).

Choosing α : Too large α can cause overshooting and divergence; too small α makes progress slow. Adaptive methods (e.g., Adam, used in neural networks) adjust α automatically.

4 Portfolio Optimization

4.1 Setup

Suppose we have n assets. Let w_i be the **weight** (fraction of wealth) invested in asset i , so $\sum_{i=1}^n w_i = 1$. Let μ_i be the expected return of asset i , and σ_{ij} the covariance between assets i and j .

Portfolio expected return:

$$\mu_P = \sum_{i=1}^n w_i \mu_i = \mathbf{w}^\top \boldsymbol{\mu}$$

Portfolio variance (risk):

$$\sigma_P^2 = \sum_{i,j=1}^n w_i w_j \sigma_{ij} = \mathbf{w}^\top \Sigma \mathbf{w}$$

where Σ is the covariance matrix.

4.2 Mean-Variance Optimization

Goal: Maximize expected return while penalizing risk. We solve

$$\max_{\mathbf{w}} \left\{ \mathbf{w}^\top \boldsymbol{\mu} - \lambda \mathbf{w}^\top \Sigma \mathbf{w} \right\} \quad \text{subject to} \quad \sum_{i=1}^n w_i = 1$$

Here $\lambda \geq 0$ is the **risk aversion** parameter: larger λ means we penalize risk more.

Unconstrained formulation: For two assets with $w_1 = w$ and $w_2 = 1 - w$:

$$f(w) = w\mu_1 + (1-w)\mu_2 - \lambda(w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12})$$

4.3 Worked Example: Two Assets

Let $\mu_1 = 0.10$, $\mu_2 = 0.05$, $\sigma_1^2 = 0.04$, $\sigma_2^2 = 0.01$, $\sigma_{12} = 0.01$.

Objective:

$$f(w) = 0.10w + 0.05(1-w) - \lambda(0.04w^2 + 0.01(1-w)^2 + 0.02w(1-w))$$

Simplifying: $f(w) = 0.05 + 0.05w - \lambda(0.03w^2 - 0.02w + 0.01)$.

First-order condition:

$$f'(w) = 0.05 - \lambda(0.06w - 0.02) = 0 \quad \Rightarrow \quad w^* = \frac{0.05/\lambda + 0.02}{0.06}$$

For $\lambda = 1$: $w^* = \frac{0.07}{0.06} \approx 1.17$. This exceeds 1—the unconstrained optimum is outside $[0, 1]$. We need the constraint $0 \leq w \leq 1$. For $\lambda = 1$, the derivative at $w = 1$ is $0.05 - 0.04 = 0.01 > 0$, so we would increase w ; the constrained optimum is $w^* = 1$ (all in asset 1) when λ is small.

For $\lambda = 5$: $w^* = \frac{0.05/5 + 0.02}{0.06} = \frac{0.03}{0.06} = 0.5$. So we hold 50% in each asset.

Interpretation: Higher risk aversion λ pushes the portfolio toward the lower-risk asset. The optimal w balances expected return and variance.

4.4 General Case: Lagrange Multipliers

For n assets with constraint $\sum w_i = 1$, we use a **Lagrange multiplier** v :

$$\mathcal{L}(\mathbf{w}, v) = \mathbf{w}^\top \boldsymbol{\mu} - \lambda \mathbf{w}^\top \Sigma \mathbf{w} + v \left(\sum_{i=1}^n w_i - 1 \right)$$

Setting $\frac{\partial \mathcal{L}}{\partial w_i} = 0$ and $\frac{\partial \mathcal{L}}{\partial v} = 0$ gives a linear system. When Σ is invertible, the solution has a closed form (Markowitz optimal portfolio).

5 Neural Network Optimization

5.1 Setup

A simple neural network: input $\mathbf{x}$, weights $\mathbf{w}$, bias b , output $\hat{y} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$ where σ is the sigmoid. We have data $(\mathbf{x}^{(i)}, y^{(i)})$ and want to choose $\mathbf{w}, b$ to minimize the error between $\hat{y}$ and y .

5.2 Loss Functions

Mean squared error (MSE):

$$L(\mathbf{w}, b) = \frac{1}{N} \sum_{i=1}^N (\hat{y}^{(i)} - y^{(i)})^2$$

Cross-entropy (for classification):

$$L(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \ln \hat{y}^{(i)} + (1 - y^{(i)}) \ln(1 - \hat{y}^{(i)}) \right]$$

5.3 Gradient Descent for Neural Networks

We minimize L over $\mathbf{w}$ and b using gradient descent:

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - \alpha \nabla_{\mathbf{w}} L, \quad b^{(k+1)} = b^{(k)} - \alpha \frac{\partial L}{\partial b}$$

Backpropagation: The chain rule gives formulas for $\nabla_{\mathbf{w}} L$ and $\frac{\partial L}{\partial b}$. For a network with many layers, we compute gradients layer by layer, propagating errors backward—hence “backpropagation.”

5.4 Worked Example: Single Neuron, MSE Loss

Let $\hat{y} = \sigma(wx + b) = \frac{1}{1 + e^{-(wx+b)}}$. For one data point (x, y) :

$$L(w, b) = \frac{1}{2} (\hat{y} - y)^2$$

Chain rule: Let $z = wx + b$. Then $\hat{y} = \sigma(z)$, and $\sigma'(z) = \sigma(z)(1 - \sigma(z)) = \hat{y}(1 - \hat{y})$.

Gradients:

$$\begin{aligned}\frac{\partial L}{\partial \hat{y}} &= \hat{y} - y \\ \frac{\partial L}{\partial z} &= \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}) \\ \frac{\partial L}{\partial w} &= \frac{\partial L}{\partial z} \cdot x = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}) \cdot x \\ \frac{\partial L}{\partial b} &= \frac{\partial L}{\partial z} \cdot 1 = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y})\end{aligned}$$

Numerical example: Suppose $x = 2$, $y = 1$, $w = 0.5$, $b = 0$. Then $z = 1$, $\hat{y} = \sigma(1) \approx 0.731$. So

$$\frac{\partial L}{\partial w} \approx (0.731 - 1) \cdot 0.731 \cdot 0.269 \cdot 2 \approx -0.105, \quad \frac{\partial L}{\partial b} \approx -0.053$$

Gradient descent updates: $w \leftarrow w - \alpha \frac{\partial L}{\partial w}$, $b \leftarrow b - \alpha \frac{\partial L}{\partial b}$. With $\alpha = 0.1$, we get $w \approx 0.551$, $b \approx 0.005$, moving $\hat{y}$ toward $y = 1$.

6 Summary

6.1 Key Concepts

1. **Critical points:** $\nabla f = 0$.
2. **Second derivative test:** Use the Hessian determinant D to classify (min, max, saddle).
3. **Gradient descent:** $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \alpha \nabla f(\mathbf{x}^{(k)})$.
4. **Portfolio optimization:** Maximize $\mathbf{w}^\top \boldsymbol{\mu} - \lambda \mathbf{w}^\top \Sigma \mathbf{w}$ subject to $\sum w_i = 1$.
5. **Neural networks:** Minimize loss L (MSE or cross-entropy) over weights using gradient descent; gradients computed via backpropagation.

6.2 Practice Exercises

Exercise 1: Find and classify all critical points of $f(x, y) = x^2 - xy + y^2 - 2x + y$.

Exercise 2: For $f(x, y) = x^3 - 3xy + y^2$, find the critical points and use the second derivative test to classify them.

Exercise 3: In the two-asset portfolio model with $\mu_1 = 0.08$, $\mu_2 = 0.04$, $\sigma_1^2 = 0.09$, $\sigma_2^2 = 0.01$, $\sigma_{12} = 0$, find the optimal weight w (fraction in asset 1) for $\lambda = 2$, assuming $0 \leq w \leq 1$.

Exercise 4: For a single neuron with $\hat{y} = \sigma(wx + b)$, data point $(x, y) = (1, 0)$, and current $(w, b) = (1, -0.5)$, compute $\hat{y}$, the loss $L = \frac{1}{2}(\hat{y} - y)^2$, and the gradients $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$.

Exercise 5: Perform one step of gradient descent for $f(x, y) = x^2 + 2y^2$ starting at $(1, 1)$ with $\alpha = 0.2$. What is the new point? Did f decrease?

Worked Solutions

Exercise 1 (Solution): $f(x, y) = x^2 - xy + y^2 - 2x + y$.

$$f_x = 2x - y - 2 = 0, \quad f_y = -x + 2y + 1 = 0$$

From the first: $y = 2x - 2$. Substitute into the second: $-x + 2(2x - 2) + 1 = 0 \Rightarrow 3x - 3 = 0 \Rightarrow x = 1$, so $y = 0$. Critical point: $(1, 0)$.

Second partials: $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = -1$. So $D = 2 \cdot 2 - (-1)^2 = 4 - 1 = 3 > 0$ and $f_{xx} > 0$. **Local minimum** at $(1, 0)$. $\square$

Exercise 2 (Solution): $f(x, y) = x^3 - 3xy + y^2$.

$$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2, \quad f_y = -3x + 2y = 0$$

Substitute: $-3x + 2x^2 = 0 \Rightarrow x(2x - 3) = 0 \Rightarrow x = 0$ or $x = 3/2$. So $(0, 0)$ and $(3/2, 9/4)$.

$f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = -3$. At $(0, 0)$: $D = 0 \cdot 2 - 9 = -9 < 0$ **saddle**.

At $(3/2, 9/4)$: $f_{xx} = 9 > 0$, $D = 9 \cdot 2 - 9 = 9 > 0$ **local minimum**.

$\square$

Exercise 3 (Solution): $f(w) = 0.08w + 0.04(1 - w) - 2(0.09w^2 + 0.01(1 - w)^2) = 0.04 + 0.04w - 2(0.09w^2 + 0.01 - 0.02w + 0.01w^2) = 0.04 + 0.04w - 0.02 - 0.2w^2 + 0.04w - 0.02w^2 = 0.02 + 0.08w - 0.22w^2$.

$f'(w) = 0.08 - 0.44w = 0 \Rightarrow w = 0.08/0.44 \approx 0.182$. So invest about 18% in asset 1 and 82% in asset 2. $\square$

Exercise 4 (Solution): $z = wx + b = 1 - 0.5 = 0.5$, $\hat{y} = \sigma(0.5) = 1/(1 + e^{-0.5}) \approx 0.622$. $L = \frac{1}{2}(0.622 - 0)^2 \approx 0.193$. $\frac{\partial L}{\partial w} = (\hat{y} - y)\hat{y}(1 - \hat{y})x \approx 0.622 \cdot 0.622 \cdot 0.378 \cdot 1 \approx 0.146$. $\frac{\partial L}{\partial b} \approx 0.146$ (same without the x factor).

$\square$

Exercise 5 (Solution): $\nabla f = (2x, 4y)$. At $(1, 1)$: $\nabla f = (2, 4)$. Update: $(1, 1) - 0.2(2, 4) = (1 - 0.4, 1 - 0.8) = (0.6, 0.2)$. Old $f = 1 + 2 = 3$, new $f = 0.36 + 0.08 = 0.44$. Yes, f decreased. $\square$